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Bottom-up nonempirical approach reducing search space in enzyme design guided by catalytic fields

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Abstract

Currently developed protocols of theozyme design still lead to biocatalysts with much lower catalytic activity than enzymes existing in nature, and so far the only way of improvement were the *in vitro* laboratory directed evolution (LDE) experiments. In this paper, we propose a different strategy based on 'reversed' methodology of mutation prediction. Instead of common 'top-down' approach, requiring numerous assumptions and vast computational effort, we argue for a 'bottom-up' one based on the catalytic fields derived directly from transition state and reactant complex wavefunctions. This enables direct one step determination of the general quantitative angular characteristics of optimal catalytic site and encompasses simultaneously both the transition state stabilization (TSS) and ground state destabilization (GSD) effects. We further extend static catalytic field approach by introducing library of atomic multipoles for aminoacid sidechain rotamers, which together with the catalytic field allow to determine the optimal side-chain orientations of charged aminoacids constituting elusive structure of preorganized catalytic environment. Obtained qualitative agreement with experimental LDE data for Kemp eliminase KE07 mutants validates proposed procedure yielding in addition a detailed insight into possible dynamic and epistatic effects.

Keywords

theozyme design, Kemp eliminase, catalytic field, atomic multipole, electrostatic effects, differential transition state stabilization

Introduction

In recent years, an intense research has been conducted in pursuit of harnessing the versatile protein capabilities for application in industrial biocatalysis.

The major challenge arises from the fact, that living organisms produce enzymes tailored for sustaining their own existence rather than the needs of human society. Therefore, the key to the future of rational biocatalyst engineering lays in redesign of existing enzymes and, ultimately, *de novo* design with the aid of new computational methods¹⁻⁴.

The major breakthroughs in theoretical design of enzymes for new, non-natural reactions (so-called *theozymes*) took place in recent two decades⁵⁻¹². However, after initial success the progress was stalled^{13,14}, as theozymes designed to maximize transition state stabilization (TSS) displayed rather low catalytic activity and conventional top-down models requiring numerous assumptions were not able to explain fully the role of additional mutations in second coordination sphere, introduced by LDE experiments^{13,14}. Therefore such LDE mutated theozymes constitute excellent proving ground for testing new methodologies and interpreting deficiencies of older methods. This problem was investigated by several research groups with various methods, including extensive Molecular Dynamics^{15,16}, QM/MM¹⁷⁻¹⁹, and EVB²⁰⁻²³ simulations. All previously mentioned methods belonging to Top-Down category require considering entire protein and numerous assumptions regarding protonation states, QM/MM boundaries, always arbitrary selection of force field etc. Although commonly used methods are able to explain some of the differences between designed and evolved enzymes, they are rather not suitable for efficient scan over vast combinatorial space. Therefore a method simple enough for high-throughput pre-screening and yet deeply rooted in the physics of the problem without involving any empirical factors is needed to improve existing protocols.

Catalytic field technique developed in our laboratory since 1981²⁴⁻²⁶, has been recently supplemented by Dittner and Hartke with extensive optimization techniques, allowing to generate and explore various characteristics of optimal catalytic environment^{27,28}. However,

extensive brute force optimization of all variables involved would still make theoretical design of theozyme composed of several hundreds of aminoacids extremely costly and not practical. We believe that systematic nonempirical analysis of the physical nature of interactions involved in catalysis and development of additional techniques specific for enzymes presented in this contribution could lead to rational construction of simpler models useful in theozyme design. The other rare inverse catalyst design attempts have been recently summarized in review²⁹ are mostly based on artificial intelligence or data mining. Although such methods could be very useful in automated scanning of vast chemical spaces, it would be surprising to obtain this way deeper understanding of physical phenomena responsible for extraordinary catalytic activity of enzymes, while many essential structural details are missing in currently available databases.

Recent experimental results employing Stark effects for specific band of IR spectra for mutated enzymes indicate electrostatic nature of KSI catalytic activity³⁰. This permits to monitor electric fields exerted by specific mutations on certain reactant bonds in Top-Down fashion yielding valuable but still limited information^{16,31}. This perspective is reversed here in Bottom-Up fashion by the use of catalytic field derived from reactant and transition state wavefunctions, yielding general characteristics of charge distribution of the optimal catalytic environment for every point in space outside reagents and enabling to monitor directly activation barrier lowering resulting from aminoacid mutations, sidechain rotations, rate-promoting vibrations or proton transfers in hydrogen bond chains.

Although catalytic field technique has been derived in our laboratory²⁴⁻²⁶ using theory of intermolecular interactions, it could be also applied to systems involving intramolecular interactions too. The ability to predict catalytic effects resulting from hydrogen to fluorine (H - > F) substitutions has been demonstrated for two simple model organic reactions³², which demonstrates qualitative applicability of this technique to model substrate assisted catalysis³³.

Present paper constitutes the first application of catalytic field approach to real case

intermolecular catalysis. For this purpose we chose here the artificial Kemp eliminase KE07 and its mutants⁹. Among other Kemp eliminases based on different scaffolds, this system has the largest number of mutants with gradually increasing activities. Several KE07 mutants introduced by LDE experiments, which apparently were missed in earlier theozyme designs^{13,14}, constitute suitable benchmark to examine new biocatalyst design methodologies.

The mutants obtained from LDE experiments differ mainly by number of charged aminoacids located in the second coordination sphere^{13,14}, which may change conformation of its sidechain upon docking substrate or nearby mutation^{34,35}. The importance of considering aminoacid sidechain rotamers in biocatalysis has been recently demonstrated for Kemp eliminase³⁶ and triosephosphate isomerase mutants³⁷. More general catalytic field technique^{25,26} considering simultaneously both TSS^{8,9} and Ground State Destabilization²³ combined with atomic multipole rotamer library scan allows to obtain valuable insight into active site pre-organization.

This contribution constitutes also the first attempt to test and validate catalytic field technique and apply it to explore possible mutation non-additivity, dynamic and epistatic effects in biocatalysis resulting from sidechain rotations.

Methods

Here, we present a bottom-up strategy based on a concept of catalytic fields derived from DTSS approach^{25,26}. Within this *ab initio*-based framework, activation barrier lowering is partitioned into *intrinsic* (*vacuum* in original paper²⁵) contribution of reactants and the difference of enzyme-transition state(TS) and enzyme-substrate(S) interaction energies E_{DTSS} .

Whenever DTSS approach is applied to enzyme reactions, substrate means ground state (alternatively reactant complex or Michaelis complex). In principle DTSS approach is more general and could be applied to other catalytic systems like zeolites³⁸.

Technically speaking, 'intrinsic' contribution is a barrier calculated for reaction path coordinates as they are in catalyst, but without inclusion of molecular environment in QM calculation. Despite virtual character of this reference state and DTSS energy for a single catalyst, it can be used as the reference in quantitative comparison of a sequence of mutants of the same enzyme. Provided that mutation does not change the reaction mechanism, taking the difference between mutant and wild type (WT) DTSS energies cancels out the 'intrinsic' contribution, leaving a value corresponding to the difference of reaction barriers of mutated and wild type enzyme (see Figure 1).

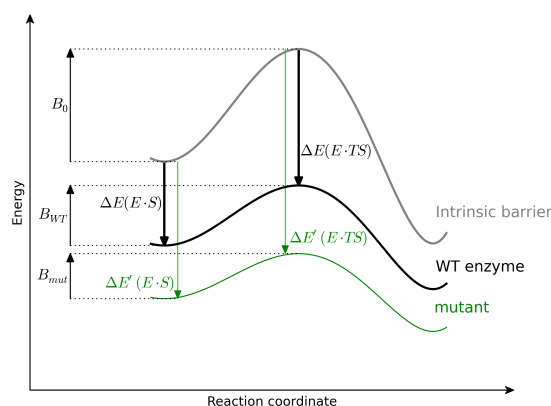


Figure 1: Differential Transition State Stabilisation (DTSS) energy $E_{DTSS} \equiv E_{IE}(TS) - E_{IE}(S)$ as a measure of catalytic efficiency; under assumption that mutation does not change reaction mechanism, the difference $E_{DTSS}^{mut} - E_{DTSS}^{WT}$ corresponds to the difference of activation barriers $B_{mut} - B_{WT}$.

$$\begin{aligned}
 E_{DTSS} &\equiv \Delta E(E \cdot TS) - \Delta E(E \cdot S) \\
 &\sim \Delta E_{EL,MTP}(E \cdot TS) - \Delta E_{EL,MTP}(E \cdot S)
 \end{aligned}
 \tag{1}$$

Furthermore, DTSS could be partitioned into the electrostatic multipole $\Delta E_{EL,MTP}$, electrostatic penetration $\Delta E_{EL,PEN}$ exchange ΔE_{EX} , delocalization (induction) ΔE_{DEL} and correlation (dispersion) ΔE_{CORR} terms defined within symmetry adapted perturbation

or hybrid variation-perturbation theories of intermolecular interactions^{39,40}. This provides not only insight into the physical nature of catalytic interactions induced upon mutation, but it allows also to derive in systematic way more approximate methods based on leading terms. Recent experimental results³⁰ and ab initio analysis of several enzyme systems^{26,41,42} indicate clearly the dominant contribution of electrostatic term ΔE_{EL} and in particular its long range multipolar electrostatic component $\Delta E_{EL,MTP}$ reflecting accurately at atomic level the molecular charge redistribution during reaction progress⁴³. Electrostatic multipolar DTSS term²⁶ constituting more general approach represents transition state stabilization (TSS)⁴⁴ and on the same time ground state destabilization (GSD)²³ effects, postulated earlier as the important contributions to enzyme catalysis. Our recent analysis of several mutants of ketosteroid isomerase⁴⁵ validated the utility of simple model based on cumulative atomic multipole expansion (CAMP)⁴⁶ in agreement with recent experimental data³⁰ confirming electrostatic nature of catalytic activity.

Catalytic fields

Further consideration of electrostatic component of DTSS energy leads to the static catalytic field $\Delta_S(\vec{r}_i)$, defined^{25,26} as a difference between electrostatic potentials $V(\vec{r}_i)$ of TS and substrate S (see equation 1). Its value denotes activation barrier change resulting from the presence of unit point charge in any location of catalytic site outside reactants. Due to additive nature of electrostatic interactions, catalytic field values obtained in one computational step constitute the solution of inverse catalysis problem in the form of $\Delta_S(\vec{r}_i)$, i.e. complementary charge distribution of optimal molecular environment $q_i(\vec{r}_i)$ ^{25,26}.

$$E_{DTSS} \simeq \sum_i q_i(\vec{r}_i)(V^{TS} - V^S)(\vec{r}_i) \simeq - \sum_i q_i(\vec{r}_i)\Delta_S(\vec{r}_i) \quad (2)$$

Catalytic field may thus serve as a general guide for rational selection of mutations or

sidechain orientations or rate-promoting vibrations by providing clues about optimal static and dynamic charge distribution of catalytic environment reducing activation barrier. The values of $\Delta_S(\vec{r}_i)$ could be rapidly calculated from CAMM derived directly from transition state and substrate quantum-chemical wavefunctions⁴⁶. Therefore calculations involving catalytic field scale as $O(N_R \cdot N_A \cdot N_{rot})$ only, where N_R and N_A indicate the number of atoms in reactants and considered mutated amino acid residue, respectively. The N_{rot} denotes average number of rotamers per mutation site (typically a few dozens). In the case when charged aminoacid is represented by single unit charge located at the charge gravity center (for example located near terminal sidechain atom of lysine or arginine), scaling could be further reduced to $O(N_R \cdot N_{rot})$, which has to be calculated only once for mutation site. The validity of such approximation is illustrated in Fig. 2.

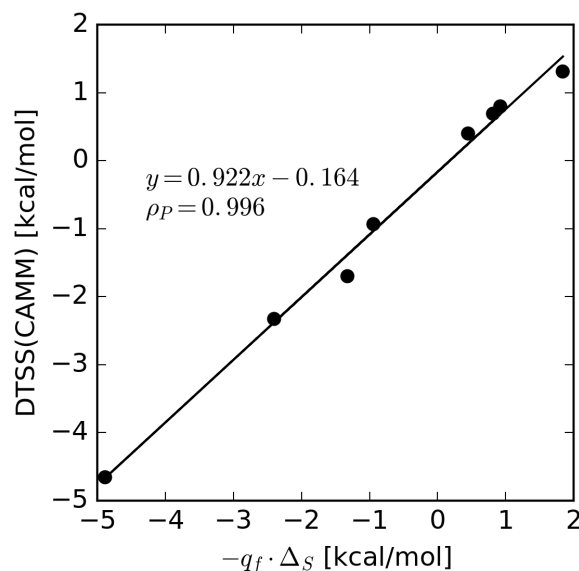


Figure 2: Activation energy changes obtained using cumulative atomic multipole moments DTSS(CAMM) versus products of catalytic fields Δ_S and formal entire aminoacid charges $-q_f$ located at its sidechain charge gravity center(Arg- guanidyl carbon, Lys- 6N , Asp- C_γ , Glu- C_δ).

This way the search space in biocatalyst design is considerably reduced within purely nonempirical approach. Use of atomic multipoles is essential to represent properly the anisotropy of molecular charge distribution at atomic level, in particular for transition states,

where bonds are formed or broken⁴³. In addition, always arbitrary atomic charge definitions are compensated by the use of higher order atomic multipoles reducing considerably basis set dependency of atomic charges⁴⁶.

It is worth to mention that Δ_S gradient field defined as dynamic catalytic field Δ_D , indicates direction of the favorable unit probe charge $q_i(\vec{r}_i)$ movement coupled with the reaction coordinate and resulting in additional decrease of the activation barrier⁴⁷. Such coupling may involve proton dislocations in hydrogen bonded networks. Recent application of dynamic electrostatic catalytic field (DECF) approach allowed us to determine that each of two steps of reaction proceeding in ketosteroid isomerase (KSI) is catalyzed by concerted proton dislocations proceeding in reverse directions in the network of 6 hydrogen bonds⁴⁸. Therefore observed exceptional KSI catalytic activity could be related to ultrafast proton switching⁴⁸.

Exploration of rotamer space

In this study we will concentrate on some unexplained mutations introduced by LDE into artificial Kemp eliminase, which are not in direct contact with substrate^{13,14}. According to aforementioned assumption about second-coordination sphere mutations (that is, their negligible effect on the reaction mechanism), the following calculations base on transition state obtained for KE07 enzyme with Gaussian09⁴⁹ ONIOM implementation⁵⁰ (details are provided in Section 2 of Supporting Information). The QM region of that simulation, consisting of 5-nitrobenzoixazole (5NBI) and acetate representing part of Glu101 residue was superimposed on 5NBI-enzyme complexes of mutants.

The space of possible amino acid conformations was explored in two ways. In the first, conventional molecular dynamics (MD) has been used with solvent effects modelled within Molecular Mechanics Poisson-Boltzmann Surface Area model MMPBSA⁵¹, which is regarded as reliable free energy simulation method to investigate protein-ligand interactions. This

provides a baseline representing a commonly used approach to the problem. In the second one, the library of atomic multipoles for aminoacid sidechain rotamers has been scanned using catalytic fields, providing a proof of concept of a proposed approach (note that with a proper implementation this involves much less computations time than MD simulation and avoids use of always arbitrary force fields).

MMPBSA approach

First, a 30 ns Molecular Dynamics trajectories were prepared for each of eight mutants, followed by MMPBSA calculation with the usage of gMMPBSA program⁵¹ Molecular Dynamics (MD) simulations were carried out in Gromacs 4.6⁵². For the sake of consistency with ONIOM calculations, Amber94 force field with TIP3 water model was used. Mutant structures were generated from KE07 by a home-made script according to published data⁹(see Table S1). All structures were submerged in water box at least 1 nm from the protein atoms, with sodium and chlorine ions for charge neutralization. In all simulations periodic boundary conditions and Particle Mesh Ewald long-range electrostatics (1 nm cut-off) were used. For MD simulations, leap-frog algorithm was used, and the temperature and pressure were set to 300 K and 1 bar, respectively. After initial steepest descent minimization, NVT and NPT equilibration stages were performed, each one 200 ps long. In this phase, modified Berendsen thermostat and Parinello-Rahman isotropic barostat were chosen. In subsequent 30 ns production run, distance restraints between Glu101 OE2 and ligand C3 and H3 atoms were used, in order to prevent random dissociation of complex. Then, trajectories were used in MMPBSA calculations done with gMMPBSA program⁵¹.

MD simulations performed for KE07 and its mutants concerned only structure of enzyme-substrate complex. It was assumed that reaction event, when takes place, is much faster than conformational changes of surrounding amino acid residues. Therefore, these simulations were supposed to provide an ensemble of protein conformations rather than to simulate

the reaction itself. Furthermore, the application of catalytic field Δ_S could be tested in this way. Taking advantage of the minor changes in geometry between reactant and TS (see Figure S3b) and keeping the assumption mentioned above, the same trajectory was assumed for TS. Different energies were obtained by substitution of charges assigned to 'QM region' (as defined in ONIOM calculations) in the topology files. Since `g_mmpbsa` evaluates electrostatic and implicit solvation contributions to binding energy (we included only electrostatic components), DTSS energy with implicit solvation was obtained by putting *differential charges* (the difference between each atomic charge in TS and reactant), which effectively are an atomic monopole representation of Δ_S . Atomic charges for RS and TS were calculated using Merz-Kollman scheme⁵³ in the same basis as used in ONIOM calculation. The contribution of QM region to solvation energy was subtracted from the result, since this method of calculation provides incorrect value of this term; on the other hand, it may be assumed to be negligible or at least constant among mutants (for more detailed derivation see Section 3 of Supplementary Information).

Scanning of CAMM library of rotamers

As the major approach in this study, we used the Dymaeonics Rotamer Database (backbone-independent, updated in 2015)⁵⁴ for preparation of library of CAMM for each amino acid rotamer.

All CAMMs were calculated with GAMESS (US)⁵⁵ with HF method and 6-311G(3d,2p) basis set. CAMMs for QM layer in both RS and TS states were obtained separately. The CAMM expansion was then used in calculation of catalytic fields and interaction energies. For each mutation, rotamers with the best and worst TSS/DTSS contribution were determined (hereafter, we will refer to the rotamer with the best DTSS contribution as optimal). Finally, we explored the configuration space resulting from combining rotamers at different sites (mathematically speaking, a cartesian product of individual site rotamer spaces in a

given mutant).

In order to test the importance of solvation effects on electrostatic interaction energies, we examine the version of library incorporating Generalized Born (GB) model, where the solvation energy is modelled according to equation (3). The Born radii were calculated using *bornRadius* software developed by Onufriev's group, utilizing an improved algorithm described in ref^{56,57}. The atomic charges for aminoacid residues were taken from Amber94 force field, whereas charges on QM region were calculated with Merz-Kollman method for HF/6-311G(3d,2p) wavefunction.

$$\Delta G_{solv} = -\frac{1}{2} \left(\frac{1}{\varepsilon_{in}} - \frac{1}{\varepsilon_{out}} \right) \sum_{i,j} \frac{q_i q_j}{f_{ij}} \quad (3)$$

$$f_{ij} = \sqrt{r_{ij}^2 + R_i R_j \exp \left(-\frac{r_{ij}^2}{4 R_i R_j} \right)}$$

Algorithms for exploration of rotamer configurational space

Since the number of rotamer combinations grows exponentially with the number of aminoacids, it is important to restrict the size search space, especially in early steps of calculation. In particular, elimination of rotamers that are unlikely to participate in final combination is of great importance. In our experiments, we found that the average number of rotamers per site is in the range 60-80. Thus, elimination of highly improbable rotamer (e.g. having short contact with the protein backbone) reduces the size of search space by $70^{(N-1)}$ (N being the number of included residues) configurations. Here, two criteria were applied: (i) geometric and (ii) energetic. The first one basically involves exclusion of rotamers which side chains have close contact (less or equal 1 Å) with protein backbone (neglecting, obviously, contacts corresponding to chemical bonds). The second one is the well-known Goldstein

criterion for singles elimination⁵⁸.

$$E_k(r_k^A) - E_k(r_k^B) + \sum_{l=1}^N \min_X (E_{kl}(r_k^A, r_l^X) - E_{kl}(r_k^B, r_l^X)) \geq \epsilon \quad (4)$$

Here, subscripts $k, l \in [1, N]$ denote position in amino acid sequence, whereas superscripts A and B indicate rotamer at particular site. The value ϵ in original work was equal to zero, however, by a choice of a positive non-zero value one can 'soften' the criterion and thus obtain larger final space. Such approach may be useful when one is interested in a set of low-energy conformers, which may be particularly desirable when considering dynamic effects.

The search algorithm, in general, can be described by following steps.

1. Loading aminoacid rotamers into specified positions in protein structure. At this stage criterion (i) is applied.
2. Calculation of interaction energy table.
3. Elimination of rotamers according to different criteria (e.g. ii).
4. Systematic search over resulting space.

Throughout presented work, we refer to the method without additional elimination criteria in point 3 as *multiscan*. The second algorithm tested, denoted as *DEE* (*Dead End Elimination*) involves criterion (ii) with negligence of rotamer self energy ($E_k(r_k^A)$) and ϵ set to either 0.0 or 0.005 hartree. The performance of both approaches was tested on P₆ mutant represented by 5 charged residues: Asp7, Glu19, Lys146, Arg202 and Asp224. A potential bottleneck may be the calculation of CAMM interaction energies (which can be several times slower than point-charge model) or loading CAMM from library (since rotation of large multipole arrays may be computationally expensive). Therefore, we compared performance of RESP model (Table S1) and CAMM (Table S2) We ensured that both approaches lead to the same rotamer configuration.

Two major observations can be drawn from presented results.

- Elimination of rotamers according to Goldstein criterion reduces the search space by 2-4 orders of magnitude, depending on assumed threshold. This in turn leads to similar reduction of time spent on scanning the rotamer space.
- Application of CAMM multipoles leads to about two-fold increase of loading time (due to required transformation of multipole arrays) and about six-fold increase of time spent on the computation of interaction energy table. In fact, this time is comparable with that required to scan whole (unreduced) rotamer space.

However, for larger sets of aminoacid residues, the problems with application of CAMM expansion can be tackled by initial reduction of search space basing on atomic charges, maybe with non-zero *epsilon*. This should allow to avoid situation when inclusion the charge anisotropy leads to different configuration.

Furthermore, there are at least two possible ways of further improvement of presented algorithm.

- Elimination of unlikely pairs from the search space (analogously to Goldstein criterion for singles).
- Further approximation of energy table. For larger rotamer sets, this could become a serious bottleneck, since its full preparation is required in all presented algorithms. However, for residues distant in space, there should be a little difference in interaction energy between their different rotamers (in other words, the rotation of sidechains should not change the interaction energy). In such a case, a whole block of interaction energy table could be filled at once.

Flowchart of computational procedure

A) Structure preparation

1. Preparation of CAMM aminoacid library (HF/6-31G*); structures of rotamers taken from Dynameonics database⁵⁴
2. Calculation of reaction path in KE07 (ONIOM B3LYP/6-311+G**:*Amber94*)
3. MD simulations of KE07 with distance restrain put on hydrogen bond between Asp and 5-nitrobenzoxazole (5NBI)
4. Clusterization of production trajectory with g-clust algorithm available in Gromacs. The main cluster was then used for analysis of conformational space

B) Exploration of rotamer space

1. Only aminoacid residues varied in directed evolution experiments⁹ were included. Rotamers from CAMM library were loaded into these positions, excluding those having short-contacts with either backbone or ligand (short-contact is defined as the shortest interatomic distance below 1.25Å)
2. Evaluation of interaction energy table in either GB model (see below) or with corresponding point charge model without solvation
3. Elimination of pairs of rotamers (of different residues) yielding short-contacts
4. Search of the lowest energy configuration with different algorithms
 - brute-force algorithm (direct iteration over all possible configurations) (referred to as *multiscan*)
 - Dead End Elimination (elimination of rotamers according to Goldstein criterion for singles).⁵⁸
 - Mean-Field algorithm.⁵⁹
5. Evaluation of DTSS(CAMM) or DTSS(GB) energy for found configuration

Results

The catalytic field, and optimal locations of crucial aminoacids

The three charged residues Asp7, Lys146 and Arg202, conserved in directed evolution⁹ (see Table S1), seem to play the most important role. What's even more remarkable, these findings are consistent with catalytic field corresponding to this reaction. As can be seen in Figure 3, charged amino acids conserved in directed evolution lay exactly where they should be located according to catalytic field Δ_S , and the range of their possible rotamers fits well corresponding 'basins'. The position of Asp7 on the edge between the two areas of Δ_S provides a straightforward exploration of both its low contribution and range varying from positive to negative (best and worst DTSS from rotamer scan are -1.1 *kcal/mol* and 0.0 *kcal/mol*, respectively).

Figure 3a presents catalytic field $\Delta_S(\vec{r}_i)$ for Kemp eliminase calculated according to equation (2) on the van der Waals surface of reactants as well as optimal rotamers of crucial charged residues.

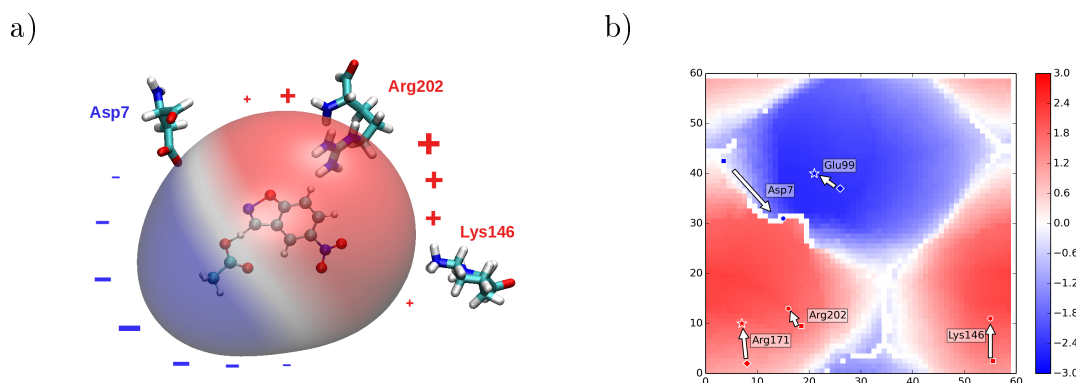


Figure 3: Catalytic field in Kemp eliminase. a) 3D view of surface colored with the value of Δ_S : red- positively charged catalyst residues will be beneficial, blue- negatively charged residues will be beneficial; b) Kohonen map⁶⁰ of the catalytic field – values in *kcal/(mol·e)*; positions of amino acids at the 7 Å distance above van der Waals surface marked with circles (optimal DTSS) and squares (the worst DTSS); residues predicted to provide further improvement marked with stars (optimum DTSS) and diamonds (the worst DTSS); the arrows indicate transition from position with the worst DTSS to the best one.

Sidechain movements and the ranges of DTSS contributions

Charged aminoacid sidechains, especially of lysines and arginines, belong to most flexible upon ligand binding^{34,35}, so it is necessary to consider entire range of their dislocations in more detail. Results obtained using library of atomic multipoles for aminoacid sidechains indicate that activation energy changes may vary between 1-4 kcal/mol for charged residues, in contrast to neutral aminoacids (0 - 0.5 kcal/mol). The possible movements of charged aminoacid terminal atoms representing extremal values of catalytic activity measured by DTSS are illustrated on 2D Kohonen map of Kemp eliminase catalytic field on Fig.3b.

In order to make a two dimensional representation of catalytic field, we used a self-organising map basing on Kohonen neural network⁶⁰. First, a surface at 7 Å distance from VDW surface of TS was generated with point density set to 25 points/Å². It was then used to train a neural network consisting of 3600 neurons arranged into a square grid with toroidal topology. This special topology means that the grid (and thus the resulting map) does not have an edge- the 'rightmost' point is topologically adjacent to the 'leftmost', while the 'topmost' neighbors the 'bottom-most'. Another consequence of this fact is that replicas of such map may be placed next to each other like tiles, showing the topological surroundings of each point. After training, the map was colored according to the value of catalytic field Δ_S , and terminal atoms of amino acids (chosen in the same manner as presented on Figure 2) were projected onto.

These sidechain conformation changes could be regarded as the important part of pre-organized catalytic environment. Relatively large activation barrier changes resulting from aminoacid sidechain movements reinforce the conclusions from previous experimental and computational studies indicating the importance of precise positioning of catalytic residues⁶¹.

Multiscan results including solvent effects - comparison with experiment

Enzyme catalytic activity could be expressed in terms of TS and S interaction energies with enzyme treated as rigid body. However, docking substrate may induce significant changes in side-chain conformation of charged aminoacids^{34,35} which may strongly interact between themselves as well as with TS and S. Whereas interaction energy for typical neutral hydrogen bonded dimer amounts 3-5 kcal/mol, it can be one or two orders of magnitude higher, if at least one of the monomers bear non-zero electric charge (specifically, the absolute value of interaction energy may reach 20 kcal/mol in the case of neutral molecule bonded to a charged one or even 100 kcal/mol when both partners are charged). Therefore the interactions involving charged aminoacids are expected to be dominant and a model relying solely on this contribution seems to be a reasonable first approximation. In the case of Kemp eliminase, 6 positions varied in LDE experiments involved charged residues: 7, 19, 123, 146, 202 and 224. Change of the charged state of these residues located beyond first coordination sphere and involved directly in reactant binding has largest chance to modulate catalytic field without affecting reactant binding pose. In addition they may strongly interact with each other and substrate binding may affect conformation of their charged sidechains. Therefore multiscan procedure seems to be a reasonable way to model these effects which could not be determined by currently available conventional experimental or theoretical methods.

For all mutants, residues at these positions have been taken into account in a search for the lowest-energy rotamer configuration using CAMM rotamer library (<http://camm.pwr.edu.pl/CAMM>).

In order to take into account solvent presence GB model has been used with selection of effective Born radii computed with *bornRadius* program developed by Onufriev and coworkers^{56,62}. With this energy function (denoted henceforth as GB), a search over rotamer space was performed in exactly same way to CAMM model. Interactions between amino acid residues were included.

Finally, we compare those results to MMPBSA⁵¹ calculations based on 30 ns MD simu-

Table 1: DTSS (CAMM) values resulting from multiscan in $[kcal/mol]$. Residue charge code: red - positive, blue - negative, black -neutral.

	Residue												
Enzyme	7		19		123		146		202		224		Total
P ₀	I	0.00 ¹	K	-0.85	Q	0.00	K	-1.36	G	0.00	N	0.00	-2.21
P ₁	I	0.00	E	0.40	R	1.75	T	0.00	R	-4.66	D	2.66	0.15
P ₂	Q	0.00	K	-0.93	Q	0.00	T	0.00	R	-4.66	D	2.26	-3.32
P ₄	D	0.02	K	-0.93	Q	0.00	K	-1.36	R	-4.66	D	2.26	-4.66
P ₅	D	-1.12	E	0.40	Q	0.00	T	0.00	R	-4.66	D	2.66	-2.80
P ₃	D	-1.12	K	-0.93	Q	0.00	E	1.19	R	-4.66	D	2.26	-3.32
P ₆	D	0.02	K	-0.93	Q	0.00	K	-1.36	R	-4.66	D	2.26	-4.66
P ₇	D	-1.12	T	0.00	Q	0.00	K	-2.13	R	-4.66	D	2.66	-5.33

1) For noncharged residues DTSS has been assumed to be zero

Table 2: Contributions of individual aminoacids to total activation barrier lowering DTSS (GB) including solvent effects estimated via generalized Born approach. Residue charge code: red - positive, blue - negative, black -neutral.

	Residue												
Enzyme	7		19		123		146		202		224		Total
P ₀	I	0.00 ¹	K	-0.12	Q	0.00	K	-0.60	G	0.00	N	0.00	-0.72
P ₁	I	0.00	E	0.04	R	0.40	T	0.00	R	-1.95	D	0.73	-0.78
P ₂	Q	0.00	K	-0.15	Q	0.00	T	0.00	R	-0.98	D	0.21	-0.92
P ₄	D	-0.24	K	-0.15	Q	0.00	K	-0.89	R	-0.70	D	0.21	-1.75
P ₅	D	-0.47	E	0.04	Q	0.00	T	0.00	R	-1.95	D	0.73	-1.65
P ₃	D	-0.24	K	-0.15	Q	0.00	E	0.33	R	-1.95	D	0.21	-1.79
P ₆	D	-0.24	K	-0.15	Q	0.00	K	-0.89	R	-0.70	D	0.21	-1.75
P ₇	D	-0.47	T	0.00	Q	0.00	K	-0.89	R	-1.95	D	0.73	-2.53

1) Assumed zero DTSS for noncharged residues

lations performed for each mutant separately (more details in section devoted to MMPBSA approach). Experimental activation free energies for Kemp eliminase mutants are compared on Fig. 4 with all three aforementioned models. What is remarkable, both models involving scan over rotamer space – that is, DTSS(CAMM) and DTSS(GB) – outperform DTSS(MMPBSA) approach in terms of the number of correctly ordered pairs N_{pred} . N_{pred} of a model⁶³ is a ratio of correctly predicted pairs to all pairs in a test set. A 'correctly predicted' or concordant pair is a pair of mutants P, Q for which $E_{DTSS}^P < E_{DTSS}^Q$ and $k_{cat}^P > k_{cat}^Q$ or $E_{DTSS}^P > E_{DTSS}^Q$ and $k_{cat}^P < k_{cat}^Q$, otherwise such a pair is counted as non-concordant. Given the total number of pairs N_{tot} in a test set consisting of N enzymes (5), the value of N_{pred} (usually expressed as a percentage) is given by equation 6. It's related to Kendall's tau τ_K by relation (7).

$$N_{tot} = \frac{N(N-1)}{2} \quad (5)$$

$$N_{pred} = \frac{N_{con}}{N_{tot}} \quad (6)$$

$$\tau_K = \frac{N_{con} - N_{non-con}}{N_{tot}} = 2N_{pred} - 1 \quad (7)$$

Whereas DTSS results obtained within GB, CAMM and MMPBSA models accounting for continuous solvent correlate with experimentally observed trends (Spearman correlation coefficients 0.92, 0.80 or 0.79 and predictivity rates 89.3, 82.1 or 78.6 percent), TSS model anticorrelates (Spearman coefficients -0.65, -0.35 or -0.67 with predictivity rates 25.0, 35.7 or 25 percent) confirming essential role of neglected ground state destabilization²³. This was probably one of the reasons of limited success of initial theozyme design^{13,14}.

Due to approximations involved, the correspondence is not linear; however, this does not disqualify the overall approach, which is intended to determine ranking or narrow the large set of possible mutations, so it would be manageable for a more sophisticated methods. Thus, it seems to be sufficient to indicate general qualitative trend, favoring DTSS over TSS model.

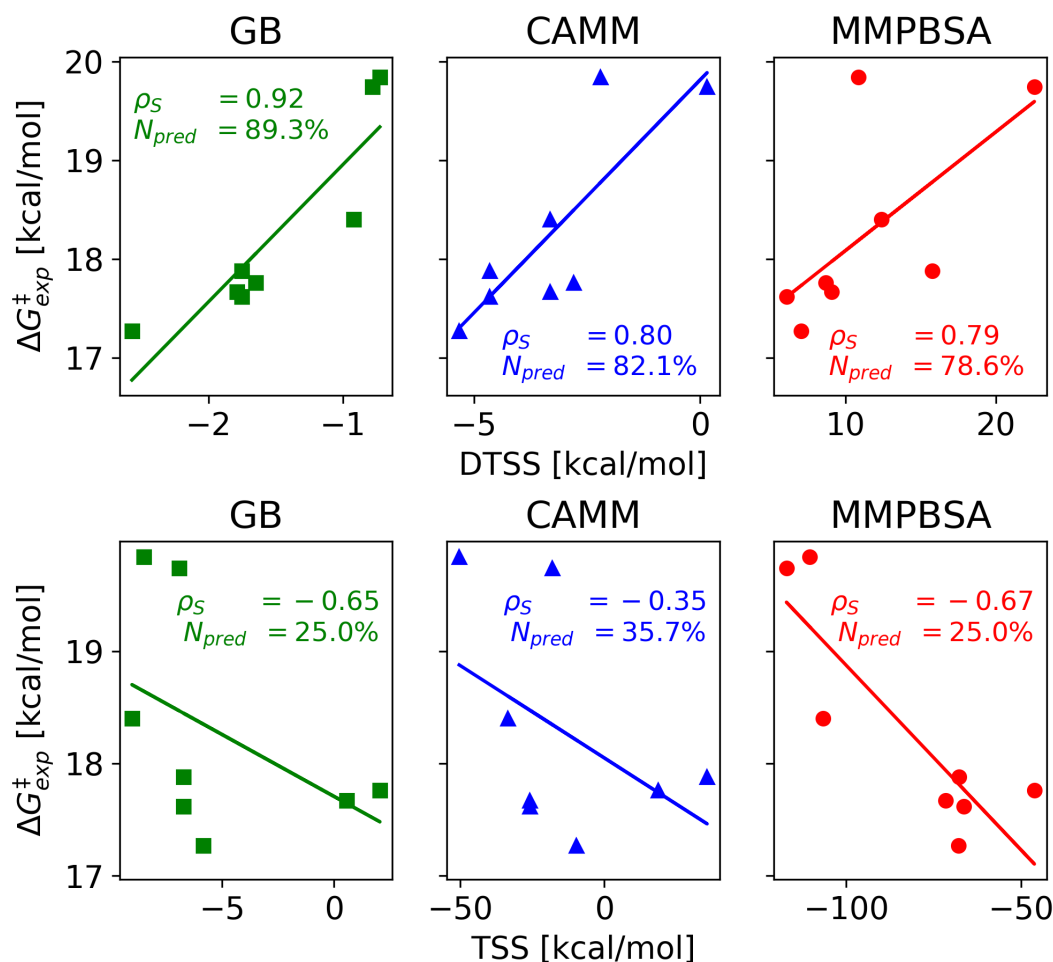


Figure 4: Comparison of theoretical DTSS and TSS results with experimental data for Kemp eliminase mutants obtained from directed evolution experiments⁸. In CAMM approach reactant-aminoacid sidechain interactions were calculated employing cumulative atomic multipole moment expansion. In GB approach corresponding ESP atomic charges have been used including solvent effects estimated from Generalized Born method. MMPBSA denote mean interaction energies using MD trajectories where solvent effects were obtained from Poisson-Boltzmann model⁵¹.

Non-additive effects of mutations

Activation barrier changes (DTSS) originating from specific residue could sometimes vary as much as 1.25 kcal/mol , indicating possible sidechain conformation change, resulting from interactions between other residues (see Table S1). Such variations responsible for nonadditivity of mutations could be examined in detail by multiscan technique presented here. Let us discuss in detail ASP7 catalytic activity in P3, P4, P5, P6 and P7 mutants. Fig. 5a-e presents schematic sidechain orientations obtained by multidimensional scaling technique preserving Euclidean distances in the space of lower dimensionality. Mutation GLU146LYS in P4 forces change of ASP7 conformation in P3 and as a result loss of its activity measured by DTSS from -1.12 to $+0.02 \text{ kcal/mol}$. Then mutations LYS19GLU and LYS146XX (XX-neutral aminoacid) introduced into P4 restore ASP7 activity in P5. In P6 previous mutation is reversed resulting in activity loss the same as in P4. Finally LYS19XX mutation in P6 restores ASP7 activity in P7 again. All these results indicate essential role of molecular context which can be explored by multiscan technique presented in previous chapter. Additional analysis of non-additive effects of LYS146 and Arg202 is presented in section 4 of Supplementary Information.

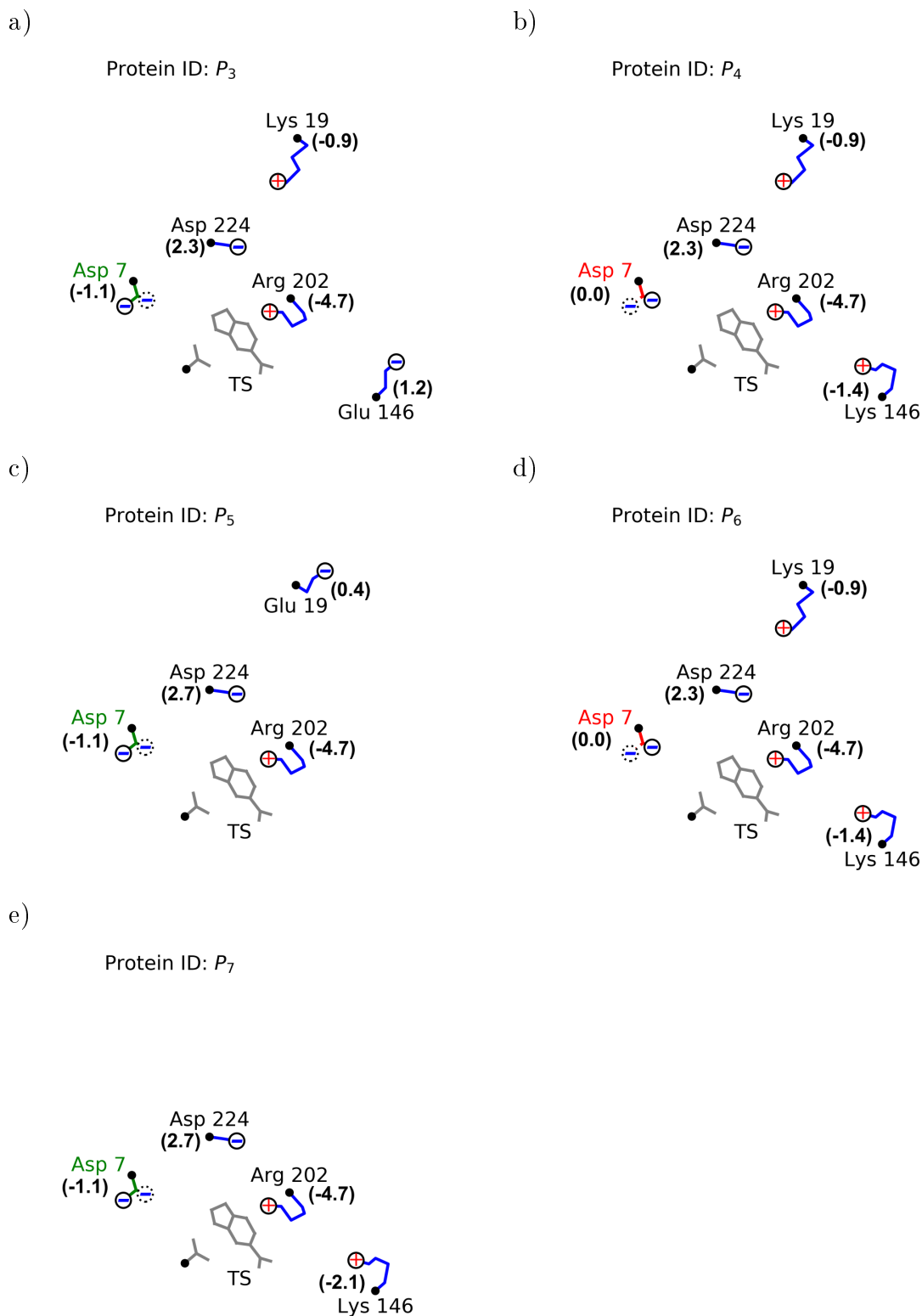


Figure 5: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in $[kcal/mol]$

Discussion

Presented scheme correctly explains the effect of mutations leading from KE07 to R7-10/11G, providing an atomistic insight into possible role of each residue contribution to the activation barrier lowering, emerging from first principles.

The most significant result of this analysis is a effective approximate solution of the 'inverse catalysis problem' confronted and confirmed for the first time by experimental results. We have shown that, basing on the information provided by catalytic field, the search space could be narrowed down to limited number of computational steps. The underlying electrostatic differential transition state stabilisation DTSS concept^{25,26} has been confirmed later in recent independent studies^{18,23,31}. Jindal and coworkers analyzed the differences between evolutionary pathways of two Kemp eliminase families: descendants of KE07 and HG3²³. They pointed out that in the first group the improvement arises from ground state destabilization, whereas in the second one from increase of Transition-State stabilisation (TSS). This is in line with our more general DTSS and catalytic field approach covering both effects simultaneously.^{25,26} Moreover, some elements of differential stabilization of transition state approach appeared later in other two works^{18,22}, which provides further reinforcement for this idea.

In summary the role of residues 7, 19, 123, 146, 202 and 224 modified in laboratory directed evolution experiments, which are not in direct contact with reactants, seems to be modulation of the catalytic field to minimize activation barrier, without changing reactant binding pose determined by primarily by aminoacids constituting first coordination sphere.

The essential advantage of present approach is the usage of cumulative atomic multipole moments to describe molecular charge redistribution during reaction progress, capturing the anisotropic character of molecular electrostatic potentials⁶⁴, which may be more significant in neutral systems⁴³. It may be still crucial in proper description of catalytic field, since Δ_S has at least dipole character due to charge conservation principle. Another important feature of catalytic field is that, despite special emphasis put on charged residues in presented

analysis, it can provide more general information; by considering its gradient $\vec{\Delta}_d$, one gets insight into optimal distribution of catalytic dipoles in a way analogous to presented above for charges (see Figures S1-S2 in Supplementary Information). Finally, as mentioned earlier, whenever necessary DTSS energy can be always generalized to include remaining interaction energy terms defined in interaction energy decomposition schemes, like SAPT³⁹ or HVPT⁴⁰. For example exchange repulsion and dispersion terms could be estimated from corresponding nonempirical atom-atom functions approximating accurate SAPT results^{65,66}. An important issue is a non-additivity of mutations reported recently²³. Within presented framework, it could be included by exploration of the effect of mutation on rotamer distribution of other residues (and corresponding DTSS energy changes). An example of such analysis is presented in Figure 5, where we found that considering mutual interactions of residues Asp7, Lys19, Lys146, Arg202 and Arg2246 in P3, P4, P5, P6 and P7 mutants changes the optimal DTSS contribution of Asp7 several times from -1.1 (P3), 0.0(P4), -1.1 (P5), 0.0 (P6) and -1.1 (P7) kcal/mol.

Such approach can be effectively used to obtain rapid estimates of nonadditive effects in multiple mutants. The general idea presented in this work may be easily introduced into other existing protocols, like an ensemble of short MD⁶⁷ or Monte Carlo simulations³⁶. / Application of static catalytic field gradients (dynamic electrostatic catalytic fields DECF)⁴⁷ allows to predict catalytic role of proton transfer in enzyme hydrogen bond networks⁴⁸.

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WCSS, ICM and PCSS.

Supporting Information

Supporting Information provides discussion regarding dynamic catalytic fields, summary of KE07 mutant data, details of ONIOM and MMPBSA simulations, as well as additional tests on non-additive mutation effects and execution times for exploration of rotamer space. CAMM rotamer library is available at <http://camm.pwr.edu.pl/CAMM>.

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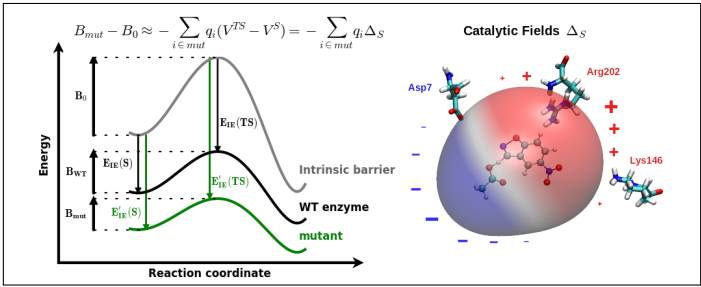
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Graphical TOC Entry



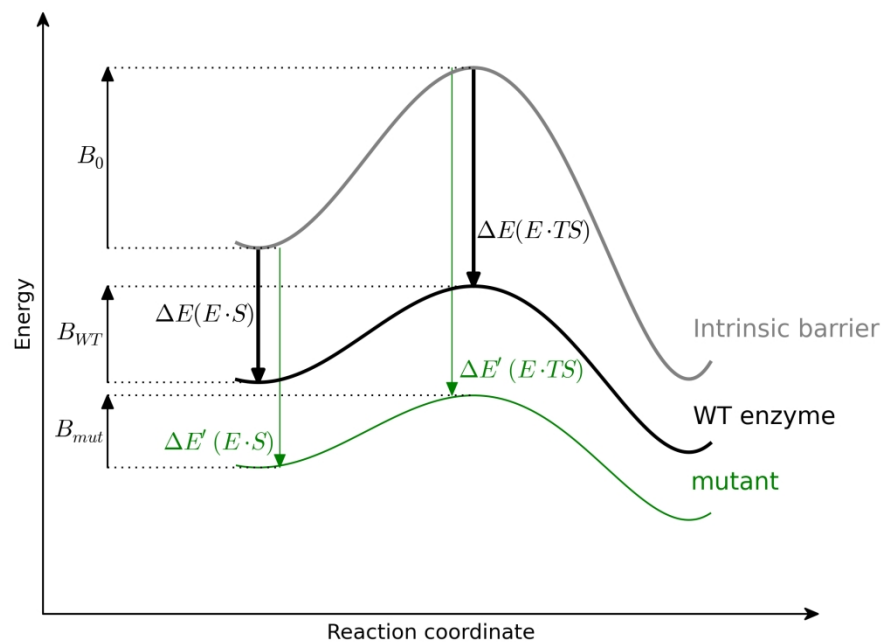


Figure 1: Differential Transition State Stabilisation (DTSS) energy $EDTSS \equiv E_{IE}(TS) - E_{IE}(S)$ as a measure of catalytic efficiency; under assumption that mutation does not change reaction mechanism, the difference $E_{DTSS}^{mut} - E_{DTSS}^{WT}$ corresponds to the difference of activation barriers $B^{mut} - B^{WT}$.

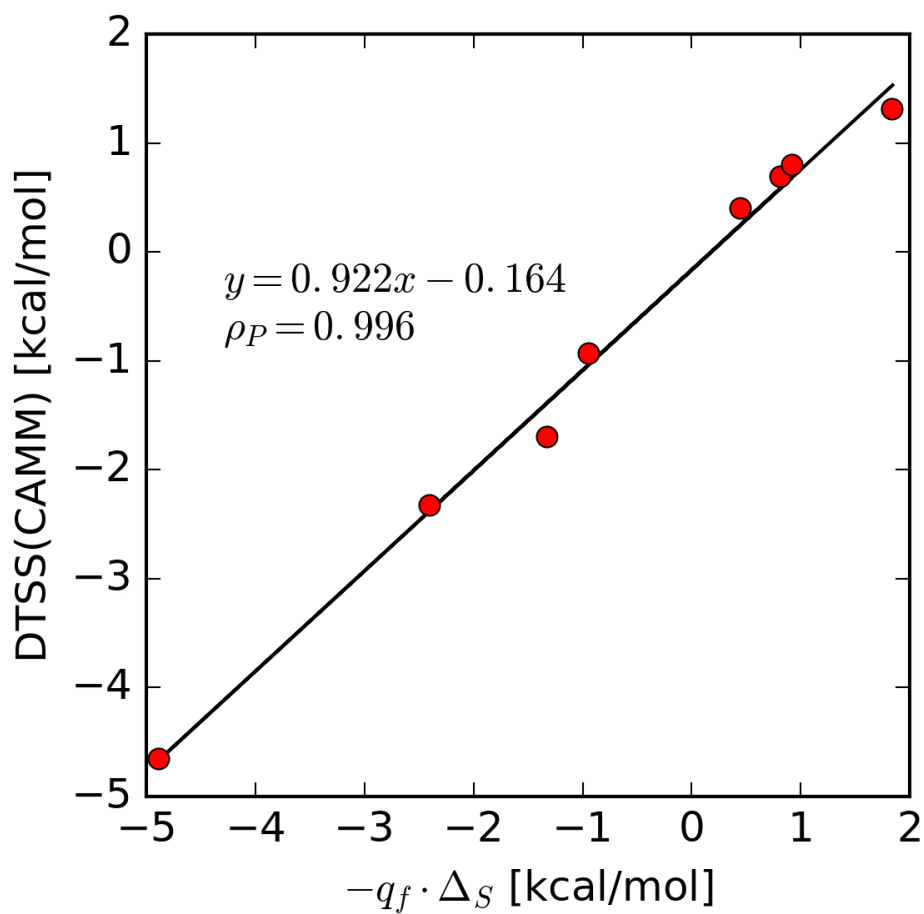


Figure 2: Activation energy changes obtained using cumulative atomic multipole moments DTSS(CAMM) versus products of catalytic fields Δ_S and formal entire aminoacid charges $-q_f$ located at its sidechain charge gravity center (Arg – guanidyl carbon, Lys – 6N, Aso – C7, Glu – C α).

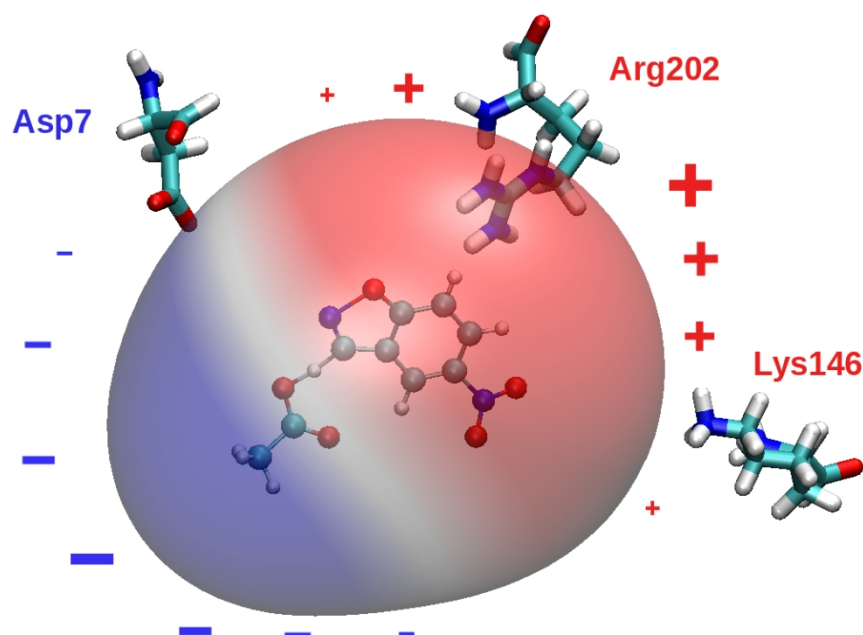


Figure 3a: Catalytic field in Kemp eliminase. a) 3D view of surface colored with the value of Δ_S : red-positively charged catalyst residues will be beneficial, blue-negatively charged residues will be beneficial; b) Kohonen map[53] of the catalytic field - values in kcal/(mol*e); positions of amino acids at the 7 Å distance above van der Waals surface marked with circles (optimal DTSS) and squares (the worst DTSS); residues predicted to provide further improvement marked with stars (optimum DTSS) and diamonds (the worst DTSS); the arrows indicate transition from position with the worst DTSS to the best one.

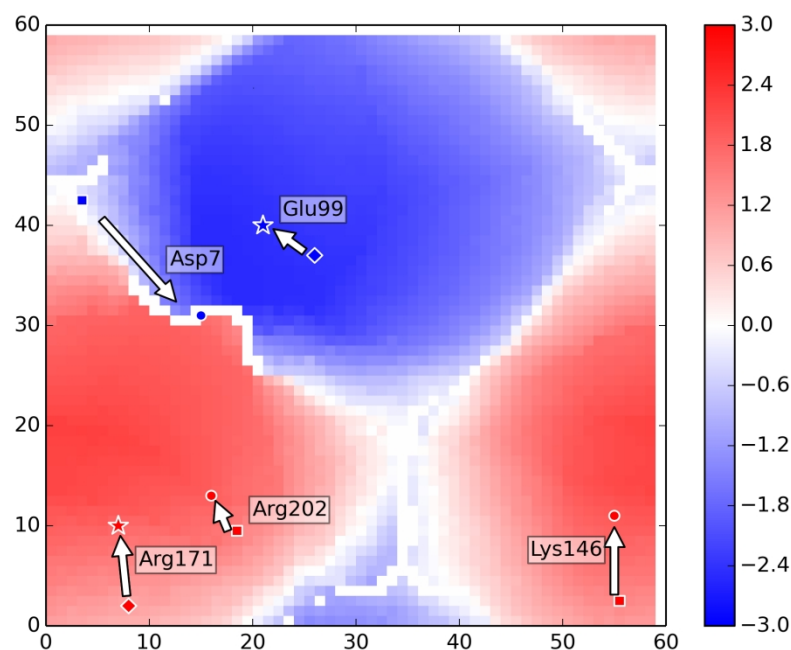


Figure 3b: Catalytic field in Kemp eliminase. a) 3D view of surface colored with the value of ΔS : red-positively charged catalyst residues will be beneficial, blue-negatively charged residues will be beneficial; b) Kohonen map[53] of the catalytic field - values in kcal/(mol*e); positions of amino acids at the 7 Å distance above van der Waals surface marked with circles (optimal DTSS) and squares (the worst DTSS); residues predicted to provide further improvement marked with stars (optimum DTSS) and diamonds (the worst DTSS); the arrows indicate transition from position with the worst DTSS to the best one.

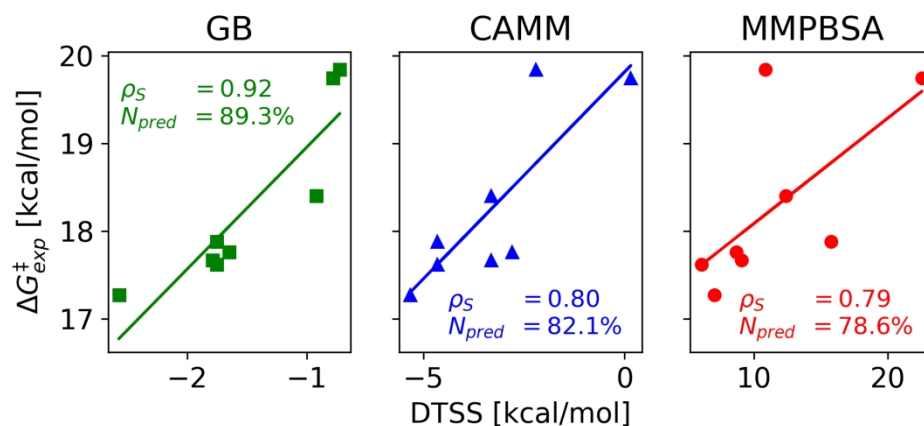


Figure 4a: Comparison of theoretical DTSS and TSS results with experimental data for Kemp eliminase mutants obtained from directed evolution experiments [8]. In CAMM approach reactant-aminoacid sidechain interactions were calculated employing cumulative atomic multipole moment expansion. In GB approach corresponding ESP atomic charges have been used including solvent effects estimated from Generalized Born method. MMPBSA denote mean interaction energies using MD trajectories where solvent effects were obtained from Poisson-Boltzmann model [44].

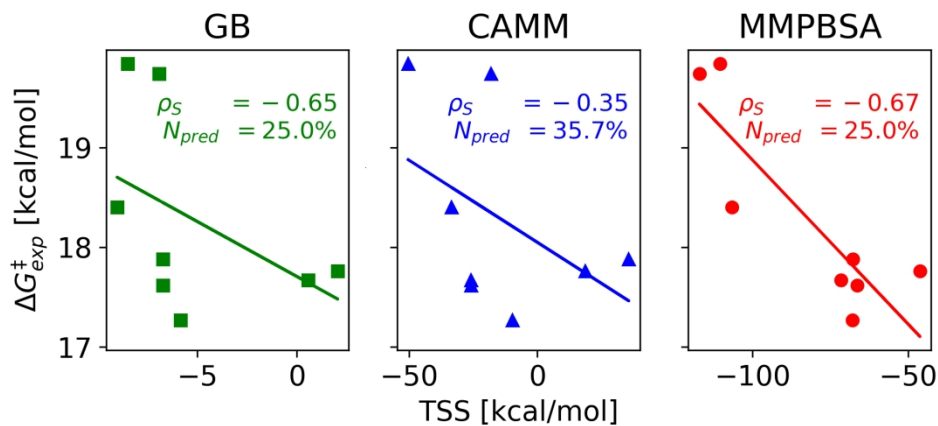


Figure 4b: Comparison of theoretical DTSS and TSS results with experimental data for Kemp eliminase mutants obtained from directed evolution experiments [8]. In CAMM approach reactant-aminoacid sidechain interactions were calculated employing cumulative atomic multipole moment expansion. In GB approach corresponding ESP atomic charges have been used including solvent effects estimated from Generalized Born method. MMPBSA denote mean interaction energies using MD trajectories where solvent effects were obtained from Poisson-Boltzmann model [44].

Protein ID: P_3

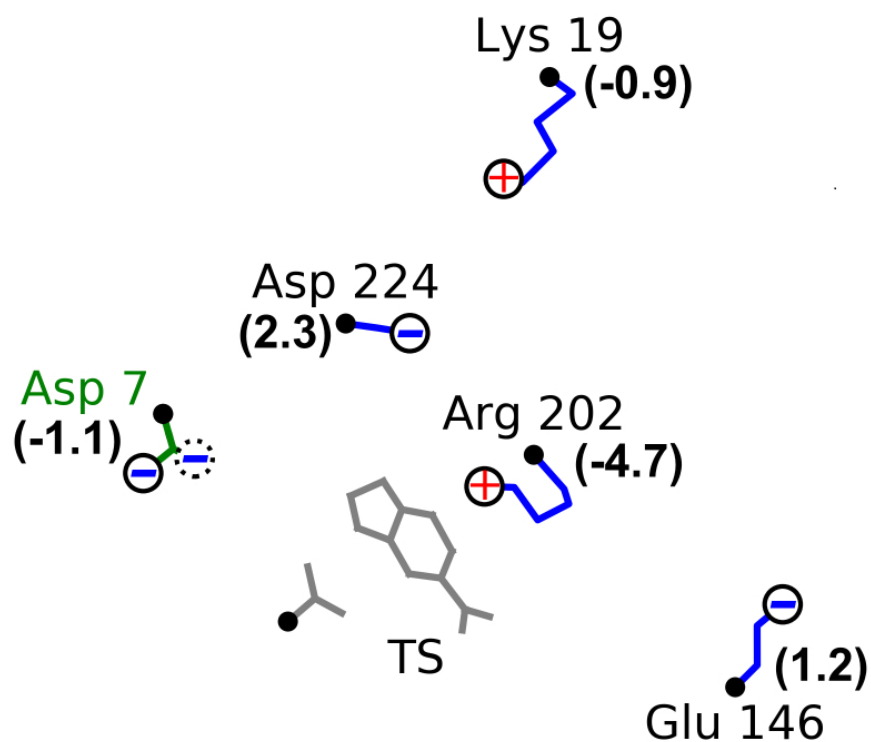


Figure 5a: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in [kcal/mol]

Protein ID: P_4

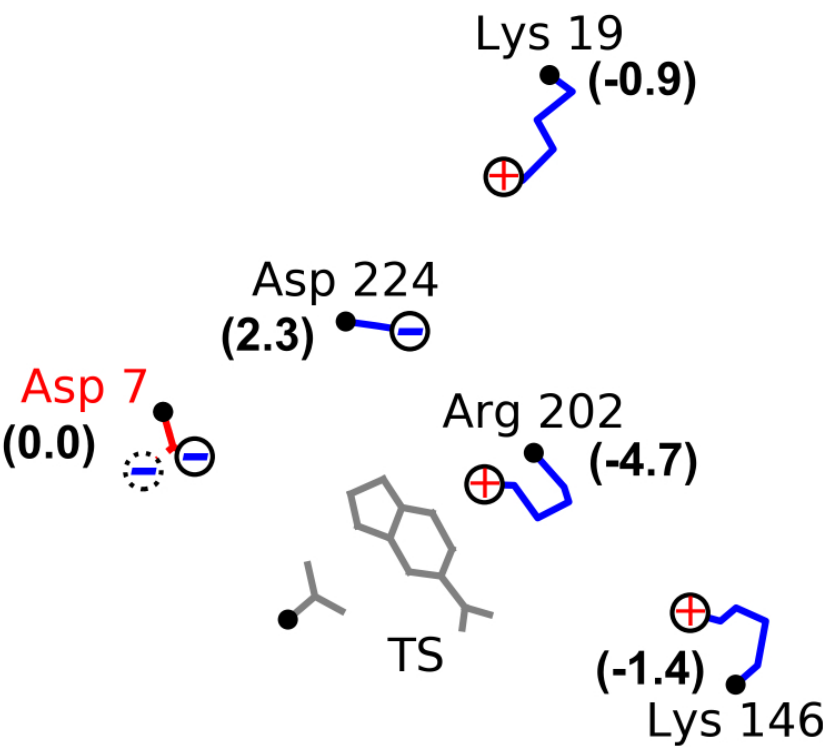


Figure 5b: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in [kcal/mol]

Protein ID: P_5

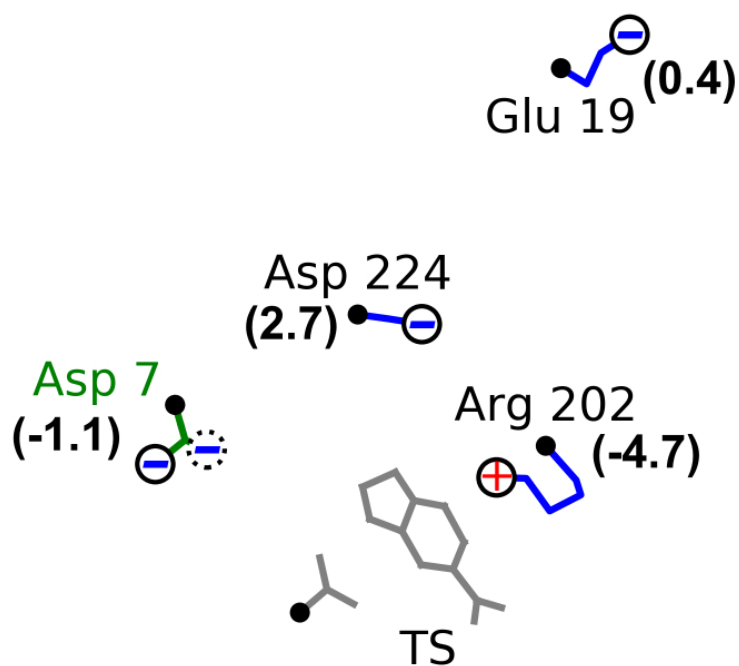


Figure 5c: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in [kcal/mol]

Protein ID: P_6

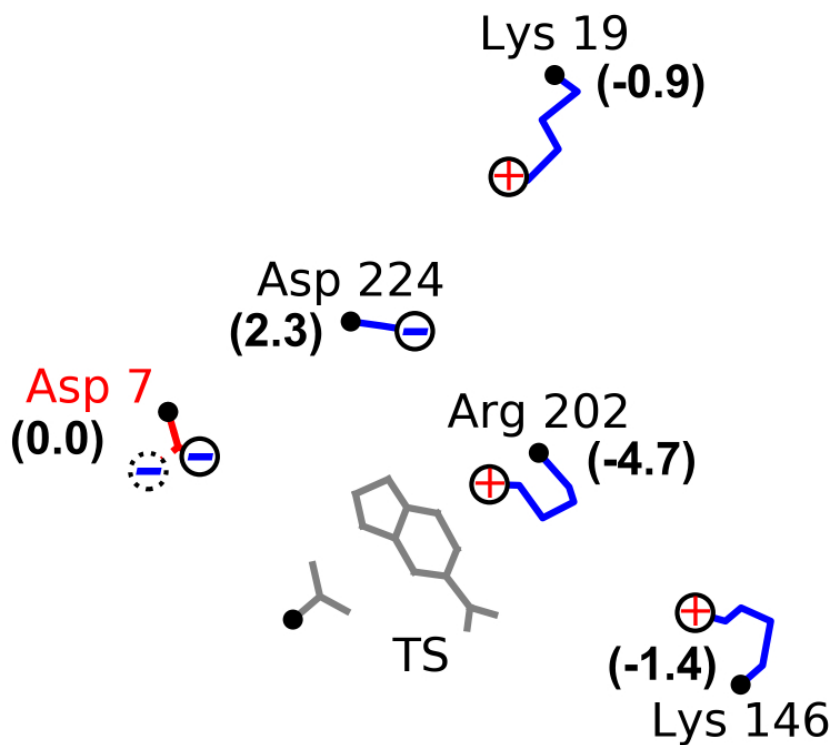


Figure 5d: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in [kcal/mol]

Protein ID: P_7

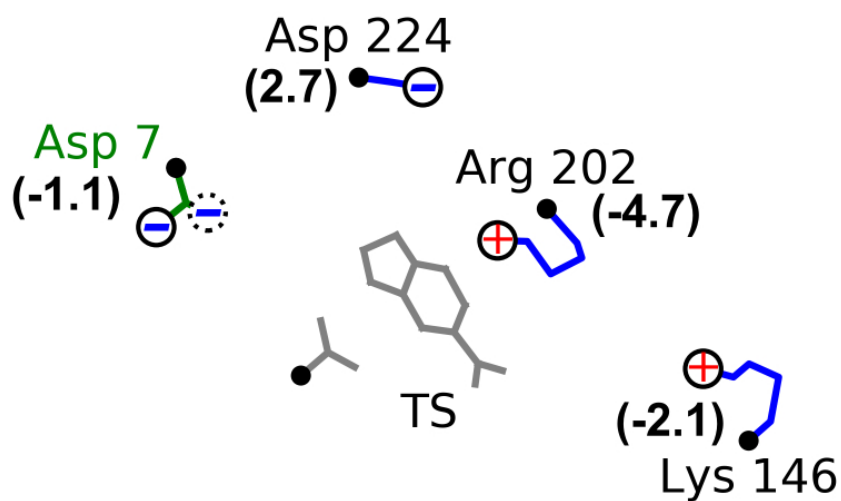
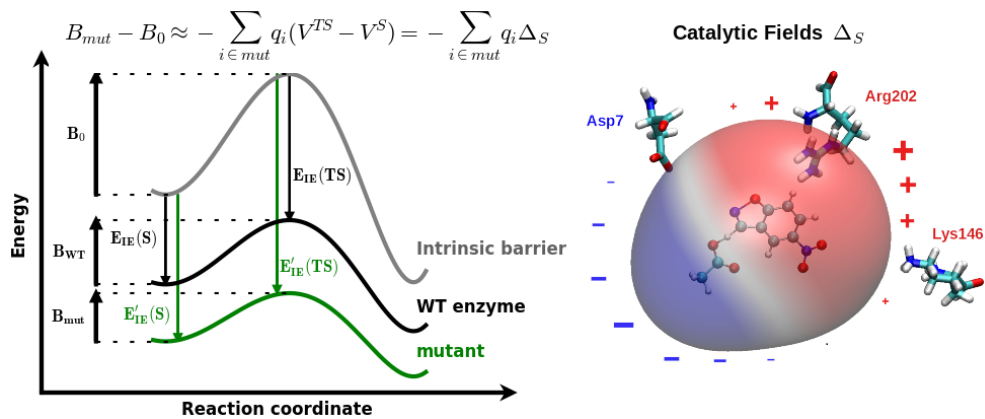


Figure 5e: Optimal sidechain orientations in Kemp eliminase active site determined for P3-P7 mutants. DTSS values in [kcal/mol]



TOC graphic abstract